

MAT 1001 Final Exam, 8:30-11:30 am, December 20, 2020

Your Name and Student ID:

Your Lecture Class(e.g, L1):

Instruction: (i) This is a closed-book and closed-notes exam; no calculators, no dictionaries and no cell phones; (ii) Show your work unless otherwise instructed—a correct answer without showing your work when required shall be given no credits; (iii) Write down ALL your work and your answers(including the answers for short questions) in the Exam Book.

1. (30 points) Short Questions (for these questions, NO need to show your work, just write down your answers in the Exam Book; NO partial credits for each question)

(i). If $f(x) > 1$ for all x and $\lim_{x \rightarrow 0} f(x)$ exists, then $\lim_{x \rightarrow 0} f(x) > 1$.

True

False

(ii). Suppose we want to approximate the positive root of $x^4 + x^2 - 1 = 0$ by using Newton's method, and we choose the first approximation x_0 between 0.25 and 0.5, then $x_0 < \text{the positive root}$ and $x_1 > x_2 > \dots > \text{the positive root}$.

True

False

(iii). Assume that $f(x)$ is continuous on $[0, 1]$, and $g(x)$ is equal to $f(x)$ at all points in $[0, 1]$, except at $x = 0, 1/n$ where $n = 1, \dots, 100$. Which of the following statement is correct? (Only one correct answer from below.)

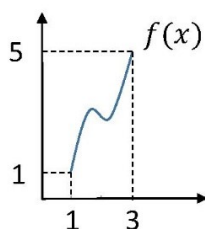
(a) g is discontinuous at $x = 0, 1/n$ where $n = 1, \dots, 100$; g is not integrable on $[0, 1]$.

(b) g is discontinuous at $x = 0, 1/n$ where $n = 1, \dots, 100$; g is integrable on $[0, 1]$ but the integrals of f and g on $[0, 1]$ may not be equal.

(c) $\int_0^1 f(x) dx = \int_0^1 g(x) dx$.

(d) There exists a function $H(x)$ such that $H'(x) = g(x)$ for all $x \in [0, 1]$.

(iv). Let $f(x)$ be a function that is differentiable on $[1, 3]$, as shown in the figure below. What's the average value of $f'(x)$ on $[1, 3]$? 2.



- (v). The improper integral $\int_{-\infty}^{\infty} \sin x dx$ is convergent and equals 0 because the integrand is an odd function.

True

False

- (vi). Let $f(x)$ be infinitely differentiable on the interval $[0, 1]$ (namely, derivatives of all orders of f exist on $[0, 1]$). Which of the following statement is incorrect?

(a) If $f(x)$ is a polynomial of order less than 4, then Simpson's rule gives the exact value of $\int_0^1 f(x) dx$. ✓

(b) If $f(x)$ is concave down on $[0, 1]$, then the midpoint rule gives an over-estimate of $\int_0^1 f(x) dx$; the trapezoidal rule an under-estimate. ✓

(c) In general, as estimates of $\int_0^1 f(x) dx$ the midpoint rule and the left sum are equally good.

(d) In general, as estimates of $\int_0^1 f(x) dx$ Simpson's rule is better than the midpoint rule. ✓

(e) The trapezoidal rule is a Riemann sum, and is the average of the left sum and the right sum. ✓

- (vii). $1 - \cos(1/x) = O(x^{-2})$ as $x \rightarrow \infty$.

$$\lim_{x \rightarrow \infty} \frac{1 - \cos \frac{1}{x}}{\frac{1}{x^2}} = \frac{\sin(\frac{1}{x}) \cdot (-\frac{1}{x^2})}{-\frac{2}{x^3}} = \frac{\sin \frac{1}{x}}{\frac{2}{x}} = \frac{1}{2} \cdot \frac{\sin \frac{1}{x}}{\frac{1}{x}} = \frac{1}{2} \text{ False}$$

- (viii). Let $p(x)$ be a polynomial whose order is bigger than 0; let $a > 1$ be a constant. Then as $x \rightarrow \infty$ we have $p(x) = o(a^x)$ and $\ln x = o(p(x))$. $\lim_{x \rightarrow \infty} \frac{\ln x}{p(x)} = 0$

$$\lim_{x \rightarrow \infty} \frac{p(x)}{2^k} = 0$$

True

$$L = \int_0^1 \sqrt{1+x} dx = \left[\frac{2}{3} (1+x)^{3/2} \right]_0^1 = \frac{2}{3} (2\sqrt{2} - 1) \text{ False}$$

- (ix). The arc length of the graph of $y = \frac{2}{3} x^{3/2}$, $x \in [0, 1]$, is _____.

- (x). A particle moves on the unit circle $x^2 + y^2 = 1$. Assume at the moment when the particle is at point $(1/\sqrt{3}, \sqrt{2}/3)$, its horizontal velocity is $10m/s$. Then at that moment its vertical velocity is $-5\sqrt{2} m/s$. $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$

2. (4 points) Match the differential equations with their slope fields.

(i) $y' = x + y$ matches ();

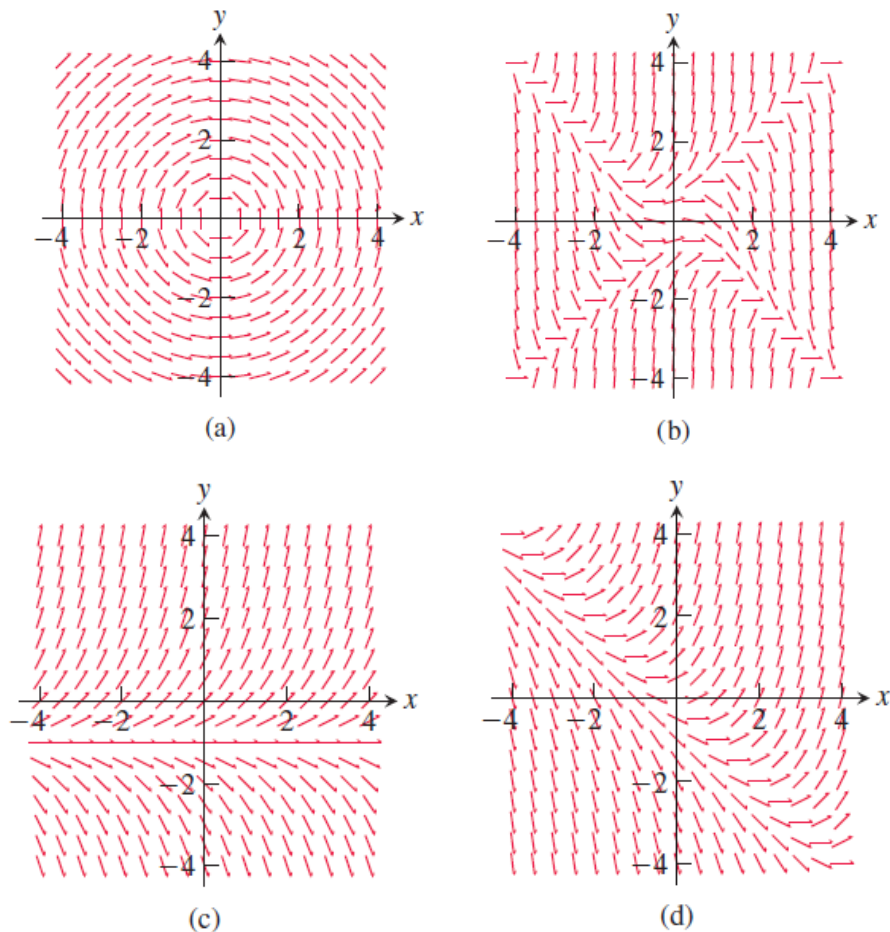
(ii) $y' = y + 1$ matches ();

$$2 \cdot \left(\frac{1}{\sqrt{3}}\right) \cdot 10 + 2 \cdot \left(\frac{\sqrt{2}}{3}\right) \cdot \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -5\sqrt{2}$$

- (i) $y' = x + y$ matches (d);
 (ii) $y' = y + 1$ matches (c);
 (iii) $y' = -\frac{x}{y}$ matches (a);
 (iv) $y' = y^2 - x^2$ matches (b).

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3. (24 points) Find each of the following limits or explain why the limit does not exist (the limit is allowed to be ∞ or $-\infty$).

(a) $\lim_{z \rightarrow 0} \frac{\sin(z^2)}{z} = \lim_{z \rightarrow 0} \frac{\sin(z^2)}{z^2} \cdot \lim_{z \rightarrow 0} z = 0$

(b) $\lim_{t \rightarrow \infty} \left(\frac{t+a}{t-a} \right)^t = \lim_{t \rightarrow \infty} \left(1 + \frac{2a}{t-a} \right)^t = \lim_{n \rightarrow \infty} \left(1 + \frac{2a}{n} \right)^{n+a} = e^{2a} \cdot \lim_{n \rightarrow \infty} \left(1 + \frac{2a}{n} \right)^a = e^{2a} \cdot 1 = e^{2a}$

(c) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{1 - \cos^2(4x)} = \frac{\tan x - \sin x}{\sin^2(4x)} = \frac{\frac{\sec^2 x - \cos x}{\cos^2 x} - \sin x}{\sin^2(4x)} = \frac{\frac{1 - \cos^3 x}{\cos^2 x}}{\sin^2(4x)} = \frac{1 - \cos^3 x}{\cos^2 x \sin^2(4x)}$
 $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{\cos^2 x \sin^2(4x)} = \frac{0}{0}$
 (d) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left(1 + \frac{i}{n} \right)^{2020} = \int_1^2 x^{2020} dx = \frac{2^{2021} - 1}{2021}$

4. (18 points) Find the derivatives.

(a) $\frac{d}{dx}(\sin x)^x = \frac{d}{dx} e^{\ln x \cdot \sin x} = (\sin x)^x \cdot \left(\frac{\sin x}{x} + \cos x \cdot \ln x \right)$

(b) $\frac{dy}{dx}$, where

$$y = \log_5 \left(\frac{\sin x \cos x}{2^x} \right) = \frac{\ln(\sin x) + \ln(\cos x) - \ln(2^x)}{\ln 5}$$

(c) Suppose that x and y are related by the equation $\frac{dy}{dx} = \frac{\cot x - \tan x - \ln 2}{\ln 5}$

$$y = \int_1^x \frac{dt}{\sqrt{10 + 3t^2}}$$

Find d^2x/dy^2 .

$$\frac{dy}{dx} = \frac{1}{\sqrt{10 + 3x^2}}$$

5. (16 points)

$$\frac{d^2y}{dx^2} = \frac{6x}{(10 + 3x^2)^{3/2}}$$

(a) Is the derivative of the following function continuous at $x=0$?

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

(b) How about the derivative of $g(x) = xf(x)$? Given reasons for your answers.

6. (8 points) Prove that

$$\rightarrow y = e^x - x - 1 \geq 0 \quad \forall x$$

for all real numbers x .

$$x=0: e^x - x - 1 = 0$$

$$y = e^x - x - 1 \geq 0$$

$$\frac{dy}{dx} = e^x - 1 \begin{cases} > 0 & \text{when } x > 0 \\ = 0 & \text{when } x = 0 \\ < 0 & \text{when } x < 0 \end{cases}$$

Loc. min at $x=0$

7. (12 points) Suppose that a differentiable function f satisfies $f(2) = -4$ and $f'(x) = \sqrt{x^2 + 5}$ for all x .

$$f'(x) = \frac{x}{2\sqrt{x^2 + 5}} > 0$$

Actual Estimate

$$= -4 + 3(-0.05) = -4.15$$

(a) Use a standard linear approximation to estimate $f(1.95)$ and $f(2.05)$.

$$-4 + 3(0.05) = -3.85$$

(b) Are your estimates in part (a) too large or too small? Explain.

8. (16 points) Consider the function $f(x) = xe^{-x}$ defined on $(-\infty, \infty)$.

Too small $\frac{d^2y}{dx^2} > 0$ for $x > 0$, so

$$= e^{-x} - xe^{-x} = e^{-x}(1-x)$$

(a) Find the intervals on which f is increasing or decreasing.

$$x < 1 \quad x > 1$$

(b) Find the intervals on which f is concave up or concave down.

$$f''(x) = -e^{-x} - e^{-x} + xe^{-x} = (x-2)e^{-x}$$

(c) Find all asymptotes (horizontal, vertical and oblique) of the graph $y = f(x)$.

$$x=0 \quad \text{nein} \quad \text{nein}$$

(d) Sketch the graph of f ; on your graph, label extreme value points, inflection points and asymptotes.



9. (12 points) State and prove Part I of Fundamental Theorem of Calculus.

Let $F(x)$ continuous on $[a, b]$, $F(x) = \int_a^x f(t) dt \rightarrow F(x)$ differentiable on $[a, b]$, $F'(x) = f(x)$

Any contin. func on $[a, b]$ always has an antiderivative

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(t) dt - \int_a^x f(t) dt}{h} = \lim_{h \rightarrow 0} \frac{\int_x^{x+h} f(t) dt}{h} = \lim_{h \rightarrow 0} f(c) \quad \exists c \in [x, x+h] = f(x)$$

MVT on def. integrals

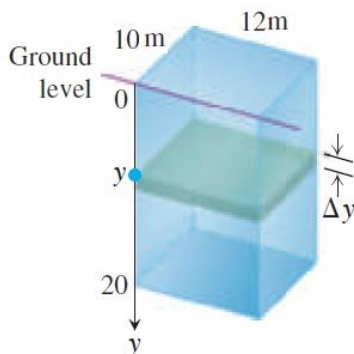
$$y = \sqrt{r^2 - x^2} \quad \frac{dy}{dx} = \frac{-x}{\sqrt{r^2 - x^2}}$$

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12. (8 points) Use Calculus to prove that the surface area of a ball is $4\pi r^2$, where r is the radius of the ball.

$$A = \int_{-r}^r 2\pi \sqrt{r^2 - \left(\frac{dy}{dx}\right)^2} dx = \int_{-r}^r 2\pi \sqrt{r^2 - x^2} \sqrt{1 + \frac{x^2}{r^2 - x^2}} dx = \int_{-r}^r 2\pi r dx = 2\pi r(2r) = 4\pi r^2$$

13. (12 points) The rectangular tank shown below, with its top at ground level, is used to catch runoff water.



Assume that the water weighs 9800 N/m^3 .

$$= \int_0^{20} 9800 \cdot 120 \cdot y \, dy$$

- (a) Set up an integral for the work needed to empty the tank by pumping the water back to ground level once the tank is full. **Do not spend your time computing the integral!**

- (b) Set up an integral for the fluid force on the side of the tank that faces you (i.e., the side with height 20m and width 12m). **Do not spend your time computing the integral!**

$$= \int_0^{20} 9800 \cdot (20-y) \cdot 12 \, dy$$

14. (12 points) Determine the convergence of each of the following improper integrals; if the integral is convergent, find its value.

(i)

$$\int_0^{\pi/2} \frac{1+e^x}{\cos x} dx. \quad u = \frac{\pi}{2} - x$$

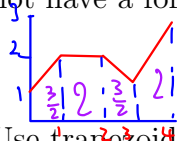
$$= \int_{\pi/2}^0 \frac{1+e^{\frac{\pi}{2}-u}}{\sin u} (-du) = \int_0^{\pi/2} \frac{1+e^{\frac{\pi}{2}}}{\sin u} du > \int_0^{\pi/2} \frac{1}{u} du \quad \text{Diverge}$$

(ii)

$$\int_1^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx. \quad u = -\sqrt{x} \Rightarrow du = -\frac{1}{2\sqrt{x}} dx$$

$$= \int_{-1}^{-\infty} -2e^u du = 2 \int_{-\infty}^{-1} e^u du = 2(e^{-1} - 0) = \frac{2}{e}$$

15. (8 points) An object is moving on a straight line with positive velocity $v(t)$. We do not have a formula for v but we have the following measurements:



$$v(0) = 1, \quad v(1) = 2, \quad v(2) = 2, \quad v(3) = 1, \quad v(4) = 3.$$

Use trapezoidal rule to estimate the total distance travelled from $t = 0$ to $t = 4$.

16. (12 pts) Solve the following differential equations

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$$e^y dy = 4x^3 dx$$

$$e^y = x^4 + C \quad y = \ln|x^4 + C|$$

(a) $\frac{dy}{dx} = 4x^3 e^{-y};$

(b) $xy' + 2y = x^3 - 1, \quad y(1) = 2.$

$$x^2 \frac{dy}{dx} + 2x \cdot y = x^4 - x \quad 2 = \frac{1}{5} - \frac{1}{2} + C \rightarrow C = \frac{23}{10}$$

$$(x^2 y)' = x^4 - x$$

$$x^2 y = \frac{x^5}{5} - \frac{x^2}{2} + C \rightarrow y = \frac{x^3}{5} - \frac{1}{2} + \frac{23}{10}$$

17. (10 points) A tank initially contains 50 liters of saltwater with concentration of salt being 0.1 kgs/liter. At time $t = 0$, saltwater of concentration 0.2 kgs/liter of salt is pumped into the tank at the rate of 5 liters/minute. Well-mixed saltwater flows out of the tank at the same rate. **Derive** the Differential Equation for the amount $S(t)$ of salt inside the tank at time t , and specify the initial condition $S(0)$. **DO NOT SOLVE the DIFFERENTIAL EQUATION.**

$$\frac{dy}{dt} = 1 - \frac{y}{50} \times 5 \rightarrow \frac{dy}{dt} + \frac{y}{10} = 1$$

18. (15 points) Compare the following two population models

$$\frac{dP}{dt} = P(100 - P), \quad (\text{Logistic Model})$$



and

$$\frac{dP}{dt} = P(P - 2)(100 - P). \quad (\text{Cubic Model})$$



- (a) Perform phase-line analysis on both models (that is, on the P -line you draw arrows to indicate the direction of motion of solutions of these two differential equations);

- (b) For each case, find $\lim_{t \rightarrow \infty} P(t)$ if $P(0) = 200$. 100 for both case

- (c) Explain, by using the results you get in (a), the difference between these two models. Why the cubic model is better than the logistic model?

If $0 < P < 2$, P will finally reach 100 in first model but reach 0 (extinct) (Too low number) in second model